



Inspiring Excellence

CSE370 : Database Systems
Project Report
Project Title : Club-Collab – University Club
Management System

Group No : 06, CSE370 Lab Section : 01, Summer 2026		
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Table of Contents

Section No	Content	Page No
1	Introduction	3
2	Project Features	4
3	ER/EER Diagram	5
4	Schema Diagram	6
5	Normalization	7
6	Frontend Development	9
7	Backend Development	11
8	Source Code Repository	13
9	Conclusion	13
10	References	14

Introduction

Club-Collab is a web-based club management system for university clubs. It helps clubs manage their members, equipment, events, and volunteers in one place.

University clubs often face these problems:

- They use paper or spreadsheets to track members
- Equipment gets lost or damaged without proper records
- Planning events and tracking volunteers is messy
- There is no easy way to recognize active members

Our solution, Club-Collab, gives clubs a single platform where they can:

- Manage members – Add, edit, and delete members. Assign roles like General Member, Volunteer, or Executive.
- Track equipment – Keep records of all equipment. Mark them as Available, In-Use, Damaged, or Maintenance.
- Log maintenance – Record repairs and costs. Equipment status updates automatically.
- Organize events – Create events with dates and venues. Track volunteer hours.
- Book equipment – Members can book available equipment for events.
- Award badges – Members earn badges automatically when they complete volunteer hours.
- Show leaderboard – See who volunteers the most.
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The system has two types of users:

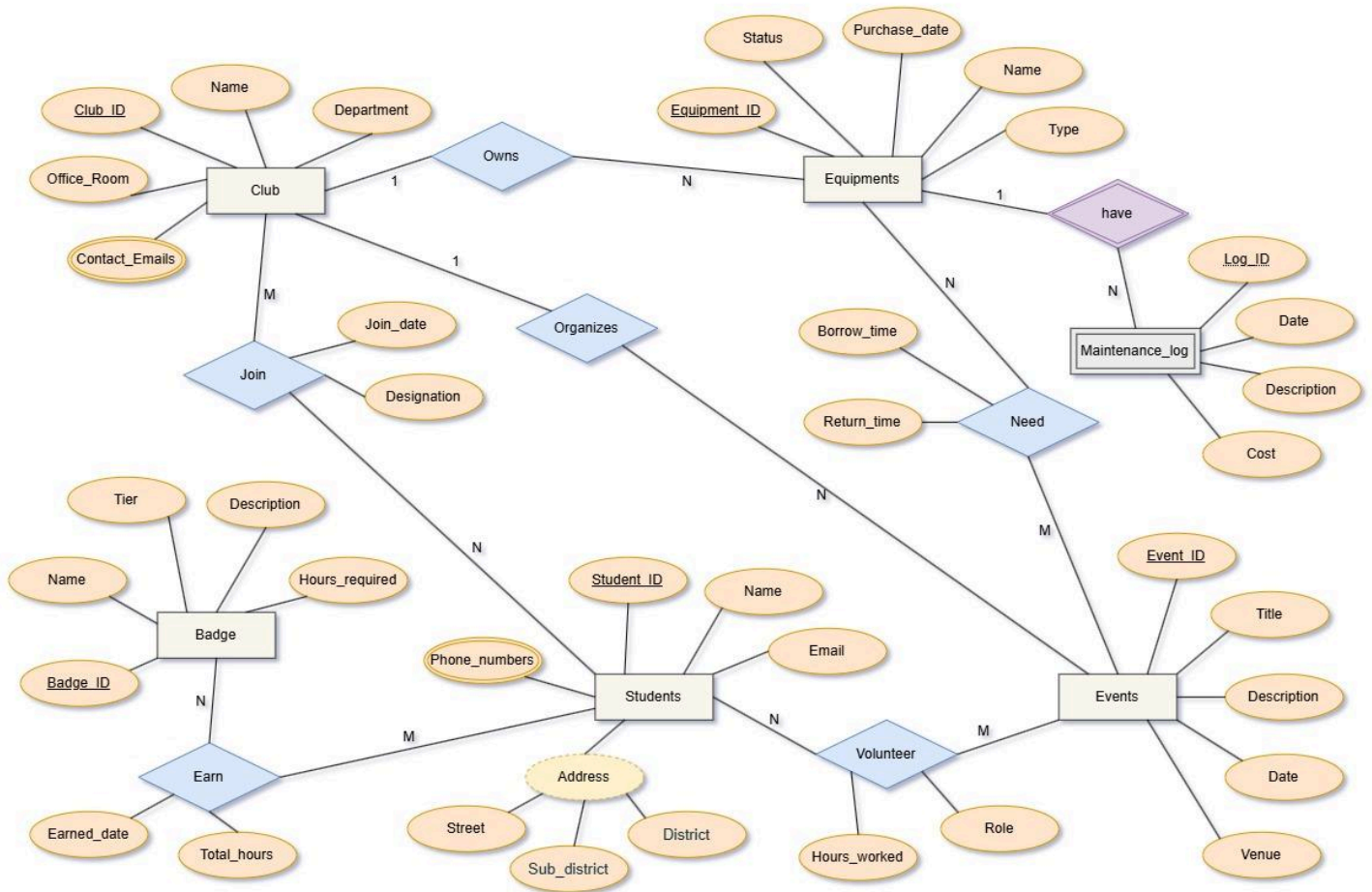
- Executives – Can add, edit, and delete anything.
- Members & Volunteers – Can only view information (Clubs, Events, Bookings, Volunteers, Badges).

We built this using MySQL for the database, PHP for the backend, and HTML, CSS, and JavaScript for the frontend.

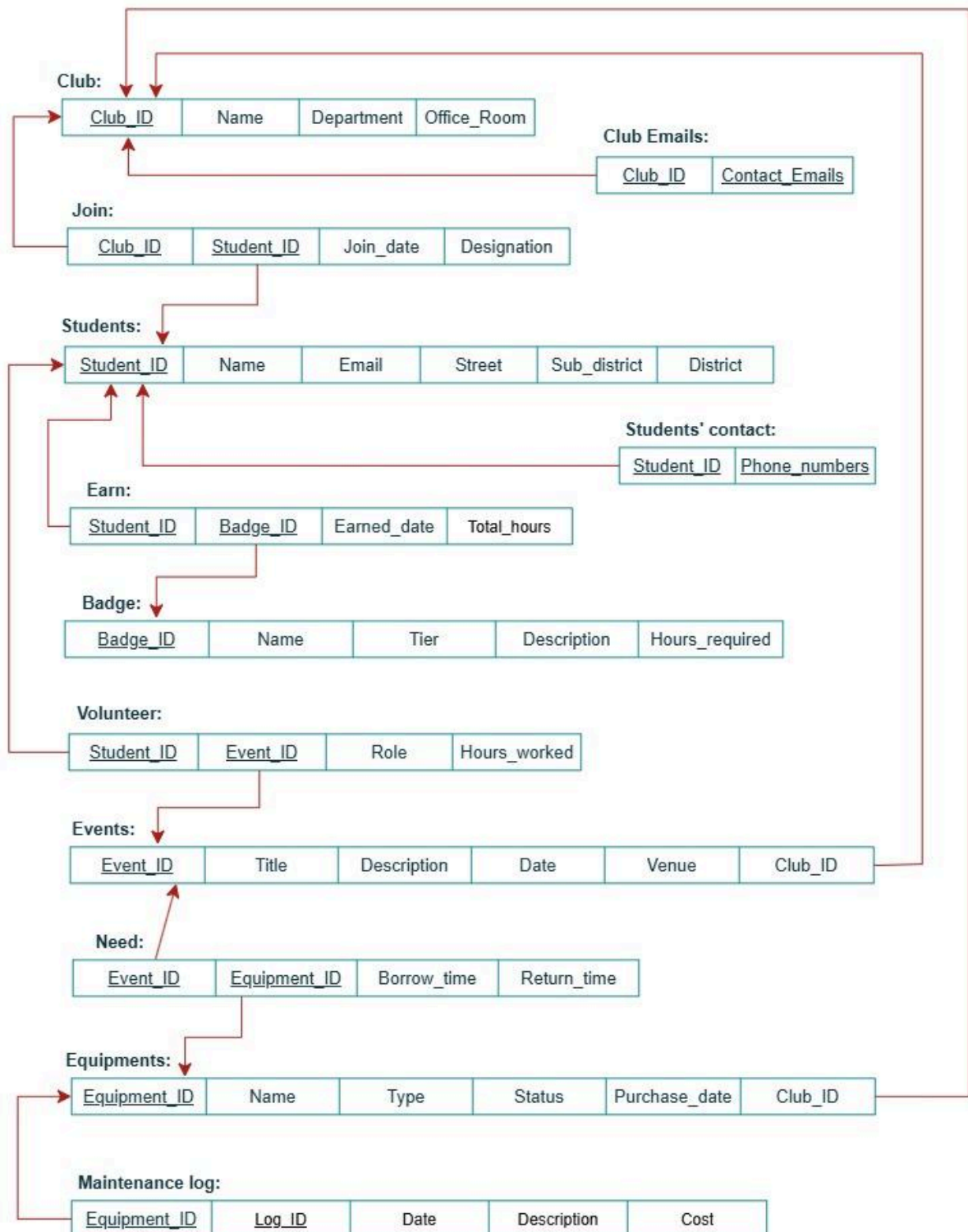
Project Features

ID, Name	Features [3 per member]	
23201662, Fahmidul Islam	Ft 1	Club & Member Management – Add, edit, and delete clubs. Add members and assign them roles (General Member, Volunteer, Executive).
	Ft 2	Event & Volunteer Management – Create events with date and venue. Log volunteer hours. Badges are awarded automatically based on total hours.
	Ft 3	Role-Based Access – Executives have full control. Members and Volunteers can only view public sections.
23231463, Saadmaan Al Naser	Ft 1	Equipment & Maintenance – Add equipment with status (Available, In-Use, Damaged, Maintenance). Log maintenance with cost. Equipment status updates automatically.
	Ft 2	Resource Booking – Book available equipment for events. Equipment becomes In-Use when booked and Available when booking is deleted.
	Ft 3	Badges & Leaderboard – Define badges with hour requirements. Auto-award badges. Show leaderboard of top volunteers.

ER/EER Diagram



Schema Diagram



Normalization

1NF (First Normal Form)

A table is in 1NF if every column has atomic (single) values.

Our Tables:

- Students – All columns have single values (Student_ID, Name, Email, etc.)
- Club – All columns have single values (Club_ID, Name, Department, etc.)
- Students_contact – Each phone number is stored in a separate row. No multiple phone numbers in one cell.
- Club_Emails – Each email is stored in a separate row. No multiple emails in one cell.
- Equipments – All columns have single values.
- Events – All columns have single values.
- Need – Borrow_Time and Return_Time are single datetime values.
- Volunteer – Role and Hours_Worked are single values.
- Maintenance_Log – Date, Description, Cost are single values.
- Badge – Name, Tier, Description, Hours_Required are single values.
- Earn – Earned_Date, Total_Hours are single values.

2NF (Second Normal Form)

A table is in 2NF if it is in 1NF AND every non-key column depends on the entire primary key (no partial dependency).

Our Tables with Composite Keys (keys made of two columns):

- Joine – Primary Key: (Student_ID, Club_ID)
Designation and Join_Date depend on BOTH Student_ID and Club_ID. A student's role is specific to a club. No partial dependency.
- Need – Primary Key: (Equip_ID, Event_ID)
Borrow_Time and Return_Time depend on BOTH equipment and event. No partial dependency.
- Volunteer – Primary Key: (Student_ID, Event_ID)
Role and Hours_Worked depend on BOTH student and event. No partial dependency.
- Earn – Primary Key: (Student_ID, Badge_ID)
Earned_Date and Total_Hours depend on BOTH student and badge. No partial dependency.
- Maintenance_Log – Primary Key: (Equip_ID, Log_ID)
Date, Description, Cost depend on BOTH equipment and log ID. No partial dependency.

Tables with Single Primary Keys:

- Students, Club, Equipments, Events, Badge – They only have one key, so partial dependency is impossible.

3NF (Third Normal Form)

A table is in 3NF if it is in 2NF AND no non-key column depends on another non-key column (no transitive dependency).

Checking Each Table:

- Students – Street, Sub_district, District depend directly on Student_ID. One address column does not decide another. No transitive dependency.
- Club – Department and Office_Room depend directly on Club_ID. No transitive dependency.
- Equipments – Name, Type, Status, Owner_Club_ID, Purchase_Date depend directly on Equip_ID. No transitive dependency.
- Events – Title, Date, Venue, Primary_Club_ID, Description depend directly on Event_ID. No transitive dependency.
- Joine – Designation, Join_Date depend directly on (Student_ID, Club_ID). No transitive dependency.
- Volunteer – Role, Hours_Worked depend directly on (Student_ID, Event_ID). No transitive dependency.
- Need – Borrow_Time, Return_Time depend directly on (Equip_ID, Event_ID). No transitive dependency.
- Maintenance_Log – Date, Description, Cost depend directly on (Equip_ID, Log_ID). No transitive dependency.
- Earn – Earned_Date, Total_Hours depend directly on (Student_ID, Badge_ID). No transitive dependency.
- Badge – Name, Tier, Description, Hours_Required depend directly on Badge_ID. No transitive dependency.

Frontend Development

Contribution of ID : 23201662, Name : Fahmidul Islam

- Built the Dashboard with KPI cards and recent activity feed.
- Built the Members and Memberships pages.
- Built the Volunteers and Events pages.
- Implemented role-based access control (hiding sections and buttons based on role).

Contribution of ID : 23231463, Name : Saadmaan Al Naser

- Built the Equipment and Maintenance pages.
- Built the Resource Booking system.
- Built the Badges and Leaderboard system.
- Designed the dark-neon theme and made it responsive.

Technologies Used

Technology	Purpose
HTML5	Page structure
CSS3	Styling and layout
JavaScript	User interactions and API calls
Font Awesome	Icons
Google Fonts	Nice typography (Space Grotesk)

What the Frontend Does

- Dashboard – Shows key numbers like total clubs, members, events, equipment, volunteer hours, and bookings. Also shows recent activity.
- Sidebar – Navigation menu. What you see depends on your role. Executives see everything. Members and Volunteers see only Clubs, Events, Bookings, Volunteers, and Badges.
- CRUD Operations – Pop-up forms for adding, editing, and deleting data. The forms change based on what you're editing (equipment, clubs, students, etc.).
- Equipment Page – Shows all equipment with status labels. You can edit status from Available to Damaged only.

- Maintenance Page – Shows maintenance history. Add maintenance for damaged equipment. When you log maintenance, equipment becomes "Maintenance" automatically.
- Booking Page – Only available equipment can be booked. When booked, equipment becomes "In-Use". When deleted, it becomes "Available" again.
- Badges Page – Shows all badges and a leaderboard of top volunteers.

Backend Development

Contribution of ID : 23201662, Name : Fahmidul Islam

- Designed the database schema.
- Built students.php, clubs.php, joine.php, events.php, volunteers.php.
- Built analytics.php for dashboard statistics.
- Implemented role-based access control backend support.

Contribution of ID : 23231463, Name : Saadmaan Al Naser

- Built equipment.php, maintenance.php, bookings.php, badges.php.
- Implemented badge recalculation and leaderboard logic.
- Implemented automatic status updates in maintenance.php and bookings.php.
- Handled auto-increment and composite key logic.

Technologies Used

Technology	Purpose
PHP	Server-side scripting
MySQL (MySQLi)	Database connection
JSON	Data format sent to frontend

What the Backend Does

The backend provides API endpoints. The frontend sends requests (GET, POST, PUT, DELETE) and the backend responds with JSON data.

Key Endpoints

Endpoint	What It Does
auth.php	Logs users in and returns their role
students.php	Add, edit, delete members
clubs.php	Add, edit, delete clubs with emails
equipment.php	Add, edit, delete equipment

maintenance.php	Add, edit, delete maintenance logs. Updates equipment status.
events.php	Add, edit, delete events
volunteers.php	Add, edit, delete volunteer logs. Recalculates badges.
bookings.php	Add, delete bookings. Updates equipment status.
joine.php	Add, edit, delete memberships
badges.php	Get badges, leaderboard, recalculate badges
analytics.php	Dashboard statistics

Some Minor Features

- Auto-increment IDs – Club_ID, Student_ID, Equip_ID, Event_ID are generated automatically by MySQL.
- Composite Keys – Some tables use two columns as a primary key:
 - Maintenance_Log (Equip_ID, Log_ID)
 - Need (Equip_ID, Event_ID)
 - Volunteer (Student_ID, Event_ID)
 - Joine (Student_ID, Club_ID)
- Automatic Status Updates – The backend updates equipment status automatically:
 - Book equipment → Equipment becomes "In-Use"
 - Delete booking → Equipment becomes "Available"
 - Log maintenance → Equipment becomes "Maintenance"
 - Delete last maintenance log → Equipment becomes "Available"
- Badge Recalculation – Every time volunteer hours change, the system checks if any new badges should be awarded.

Source Code Repository

<https://drive.google.com/drive/u/1/folders/1IWeKOPaxYbNklVuvJkgBCc3OJVfb3rhW>

Conclusion

Club-Collab is a complete club management system. It helps university clubs manage members, equipment, events, volunteers, and badges in one place.

What We Achieved:

- Clean Database – Our database is in 3NF, so there is no duplicate data and everything works efficiently.
- Automatic Updates – Equipment status updates automatically when bookings and maintenance logs are added or deleted.
- Badge System – Members earn badges automatically based on volunteer hours. This encourages participation.
- Role-Based Access – Executives have full control. Members and Volunteers can only view what they need.
- Easy to Use – The interface is clean and simple. Forms are easy to fill.
- Responsive – The website works on desktop and mobile devices.

What We Learned:

- Working on this project taught us:
- How to design a database from scratch
- How to normalize tables properly
- How to build a REST API with PHP
- How to connect frontend and backend
- How to handle user authentication and authorization
- How to manage composite keys in queries
- How to write clean, maintainable code

Possible Future Improvements:

- Email notifications for events and bookings
- Export reports to PDF or Excel
- Mobile app
- Charts and graphs for analytics

References

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